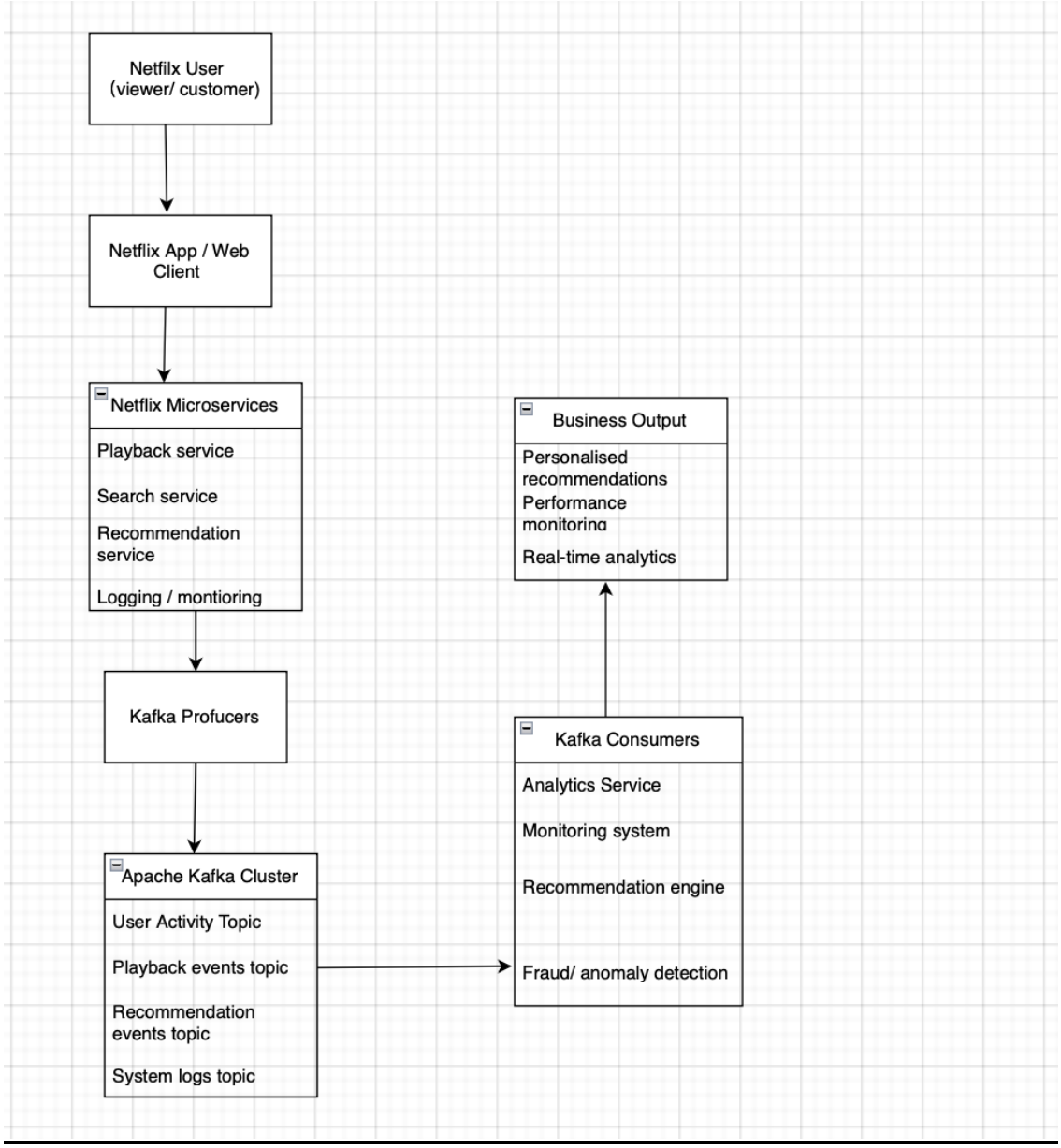
Apache Kafka Case Study – Netflix

<u>1. Identify the company that uses Apache Kafka for the system architecture</u>



One of the companies that uses Apache Kafka in its system architecture is Netflix, a global streaming platform that provides movies, television shows, and digital entertainment to millions of users worldwide. Due to the large number of users interacting with the platform simultaneously, Netflix generates massive volumes of real-time data from user activities such as video streaming, browsing content, searching for movies, and interacting with recommendation systems. To efficiently manage and process this continuous stream of data, Netflix integrates Apache Kafka as part of its large-scale data infrastructure to support high-throughput data processing and real-time event streaming (Kreps, Narkhede, & Rao, 2011; Apache Kafka, 2024).

Apache Kafka functions as a distributed event streaming platform within Netflix's microservices architecture. It enables different services to publish and consume event data in real time, allowing Netflix to collect and process user activity logs, operational metrics, and system monitoring events efficiently. This architecture allows Netflix to support large-scale data pipelines and real-time analytics while maintaining reliable streaming services for millions of users worldwide (Netflix Technology Blog, 2023; Confluent, 2023).

2. Specify the exact Apache Kafka component used in the system

In the system architecture of Netflix, several core components of Apache Kafka are used to manage real-time data streaming. One important component is the Kafka Producer, which is responsible for sending event data to Kafka topics. In Netflix's platform, various microservices act as producers, including playback services, user activity tracking systems, and application monitoring services. These services continuously generate large volumes of data whenever users watch content, pause videos, search for titles, or interact with the platform. The generated events are then categorized and stored in Kafka topics, which organize the data streams based on different types of events such as user activity logs, system monitoring data, and recommendation events (Apache Kafka, 2024; Confluent, 2023).

Another key component used in Netflix's architecture is the Kafka Broker and Kafka Consumer. Kafka brokers act as servers that store, manage, and distribute streaming data across the Kafka cluster. Netflix deploys multiple brokers to ensure high scalability, reliability, and fault tolerance when handling billions of events generated daily by its platform. Meanwhile, Kafka consumers are services or applications that read and process messages from Kafka topics. These consumers include analytics platforms, monitoring systems, and machine learning pipelines that analyze streaming data to improve system performance and generate personalized content recommendations for users (Netflix Technology Blog, 2023).

3. Discuss two advantages and disadvantages of using Apache Kafka

One major advantage of using Apache Kafka in Netflix's system architecture is its ability to support real-time data streaming and high scalability. Netflix handles massive volumes of user-generated events every second, including streaming activity, search queries, and user interactions with the platform. Kafka enables these events to be processed and transmitted in real time, allowing Netflix to monitor system performance, detect anomalies, and update recommendation systems quickly. Furthermore, Kafka's distributed architecture allows Netflix to scale its infrastructure horizontally by adding more brokers and partitions to support increasing data traffic without affecting system performance (Kreps et al., 2011; Confluent, 2023).

Despite these advantages, there are also disadvantages associated with using Apache Kafka in large-scale systems such as Netflix. One challenge is the complexity of managing Kafka

clusters, as configuring brokers, partitions, replication, and data retention policies requires specialized technical expertise. Additionally, maintaining large Kafka infrastructures requires continuous monitoring and hardware resources, which increases operational and infrastructure costs. Organizations like Netflix must allocate significant engineering resources to maintain system reliability and ensure efficient data streaming operations (Apache Kafka, 2024; Netflix Technology Blog, 2023).

4. References

Apache Kafka. (2024). *Apache Kafka documentation*. https://kafka.apache.org

Netflix Technology Blog. (2023). *Real-time data infrastructure at Netflix*. https://netflixtechblog.com

Confluent. (2023). *Apache Kafka use cases*. https://www.confluent.io

Kreps, J., Narkhede, N., & Rao, J. (2011). *Kafka: A distributed messaging system for log processing*. LinkedIn Engineering.